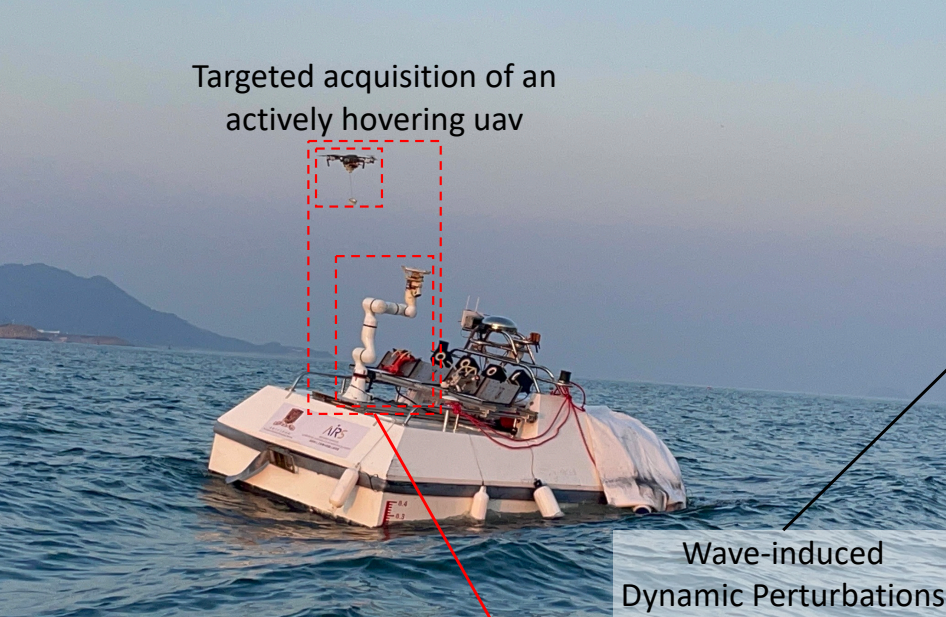
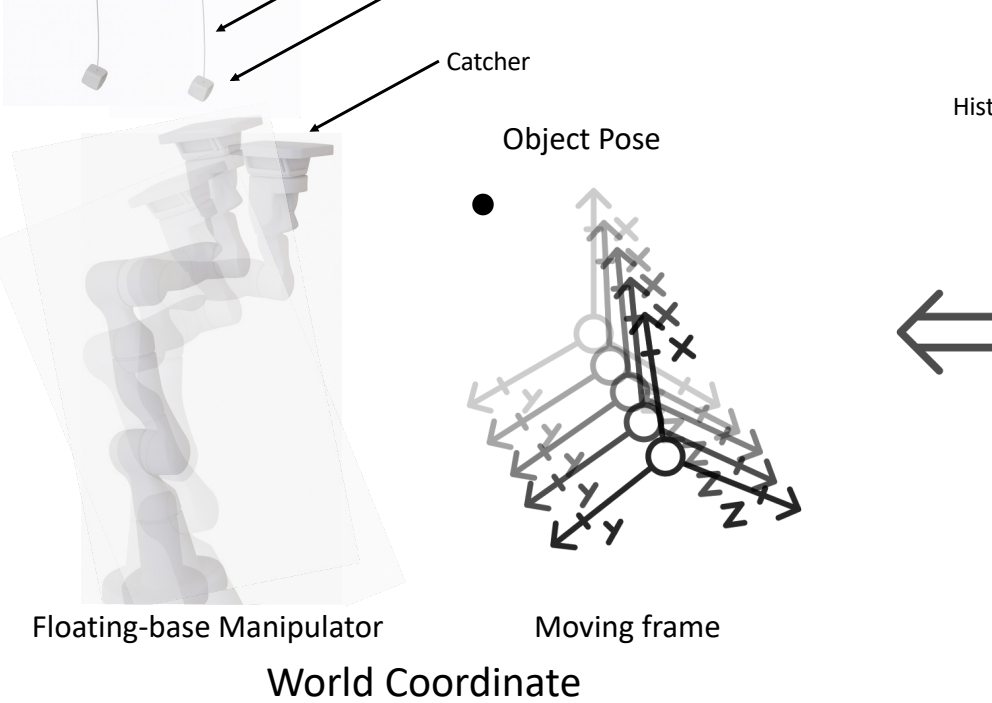


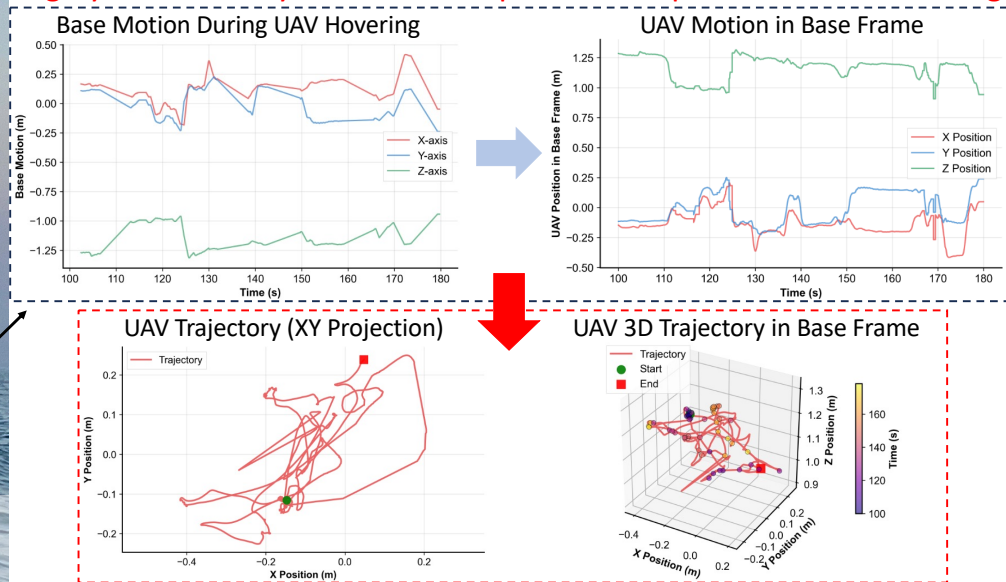


(a)

Real-Time Observed UAV Pose
Predicted UAV Pose
Tether
Docking block assembly with its iron plate base, enabling simultaneous suspended payload capture

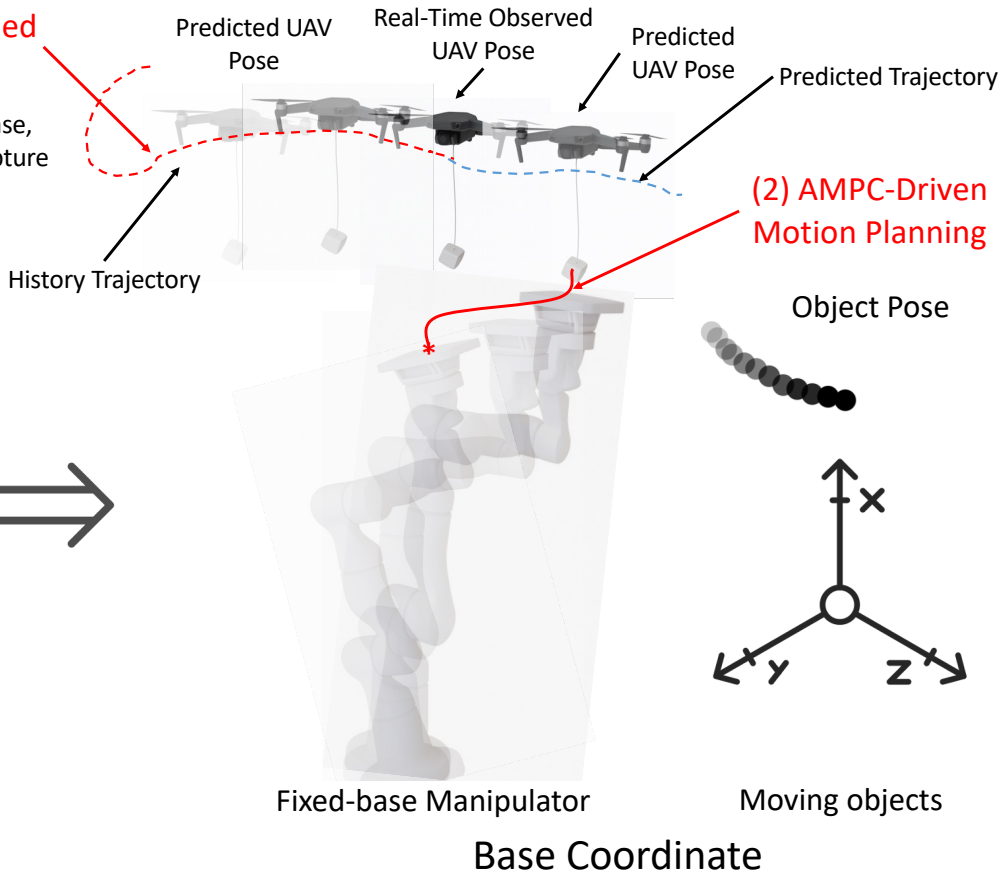


(c)



(b)

(1) KalmanNet++ Aided Motion Analysis



(d)

(2) AMPC-Driven Motion Planning